Chapter 25 worked solutions

Image and Video Generation

Exercises 1–4 work through signal and noise mixtures, a sampling step, guidance, and video attention by hand. Exercises 9–11 design checks for sampling settings, motion, and variety.

1. Mix signal and noise

Think of the clean value 2 as one pixel and -1 as the noise drawn for that pixel. The noise is negative, so its contribution lowers the result. The forward mixture scales the clean value x_0 by $\sqrt{\bar{\alpha}_t}$ and the noise ϵ by $\sqrt{1-\bar{\alpha}_t}$ before adding them.

$\bar{\alpha}_t$	Signal weight	Noise weight	Noisy value x_t
0.64	$\sqrt{0.64} = 0.8$	$\sqrt{0.36} = 0.6$	$0.8(2) + 0.6(-1) = 1.0$
0.36	$\sqrt{0.36} = 0.6$	$\sqrt{0.64} = 0.8$	$0.6(2) + 0.8(-1) = 0.4$

The weights swap. With $\bar{\alpha}_t = 0.36$, the positive signal contributes 1.2 instead of 1.6, and the negative noise contributes -0.8 instead of -0.6. Both changes move this particular mixture downward.

2. Recover a clean estimate

Start with $x_t = 1.0$ from the first mixture. The predicted noise is -0.5. Its contribution at the current step is $0.6(-0.5) = -0.3$. Subtract that contribution and divide by the signal weight 0.8 to recover the estimated clean value.

$$\widehat{x}_0 = \frac{1.0 - 0.6(-0.5)}{0.8} = 1.625.$$

The deterministic DDIM update mixes this estimate with the same predicted noise, now using the weights for $\bar{\alpha}_s = 0.81$. The signal weight is 0.9 and the noise weight is $\sqrt{0.19}$.

$$x_s = 0.9(1.625) + \sqrt{0.19}\,(-0.5) = 1.4625 - 0.5\sqrt{0.19} \approx 1.2446.$$

To see where the error goes, keep x_t fixed and write the prediction error as $\delta = \widehat{\epsilon}_t - \epsilon = -0.5 - (-1) = 0.5$. The clean estimate subtracts the weighted prediction, so replacing the true noise with the prediction changes $\widehat{x}_0$ by $-(0.6/0.8)\delta = -0.375$. That error reaches the next sample through the clean term. The next sample also uses the prediction directly in its noise term.

Part of the next sample	Change from using the predicted noise
Clean term $0.9\widehat{x}_0$	$-0.9(0.6/0.8)\delta = -0.3375$
Noise term $\sqrt{0.19}\ \widehat{\epsilon}_t$	$\sqrt{0.19}\ \delta \approx +0.2179$

The two changes partly offset each other. Together they lower the next sample by about 0.1196 relative to the exact-noise update, which gives $0.9(2) + \sqrt{0.19}\,(-1) \approx 1.3641$.

3. Change a guidance scale

Start at the unconditional point $(0, 1)$. Moving from that point to the conditional point $(2, 0)$ takes a step of $(2, -1)$. In the chapter's classifier-free guidance convention, a scale g moves g times that step from the unconditional point. The guided noise vector is therefore $(0, 1) + g(2, -1)$.

Scale g	Guided vector
0	$(0, 1)$
1	$(2, 0)$
1.5	$(3, -0.5)$

The third point lies on the same line, past the conditional point. It is one and a half steps from the unconditional point, or half a step $(1, -0.5)$ beyond the conditional point. Going past the endpoint in this way is called extrapolation.

4. Count video attention pairs

Pick the third patch in the second frame as a query. In joint attention it can read all 30 patches in the clip. In spatial attention it can read the six patches in its frame. In temporal attention it can read the third patch in each of the five frames. Each list includes the query patch itself.

There are $5 \cdot 6 = 30$ queries. Each ordered pair specifies which patch asks and which patch it reads, so two different patches can contribute a pair in each direction. The counts below are for one clip and one head in each attention operation.

Attention operation	Ordered query–key pairs
Joint across all frames and patches	$(5 \cdot 6)^2 = 900$
Spatial within each frame	$5 \cdot 6^2 = 180$
Temporal at each patch site across frames	$6 \cdot 5^2 = 150$

Running both separate passes computes $180 + 150 = 330$ scores per head across the two operations. Each pass includes its own self-pairs. These counts describe the size of the attention comparisons. Timing also depends on how many heads and how wide they are, the projections that form the queries, keys, and values, and how the hardware executes the operations and moves the data. Two separate passes can have separate projection and launch costs. Full generation also repeats attention across layers and denoising steps, so the pair counts alone cannot give a wall-clock speedup.

9. Compare two sampling settings

Suppose we ask a generator for *a red cube suspended above a blue sphere on a white background*. Save a starting noise tensor for the prompt. Generate once with 10 sampling steps and once with 30 steps, reusing that tensor. If the two images differ, the comparison has held the initial random input fixed. Repeat with several prompts and several noise tensors per prompt to see how consistently a difference appears.

Freeze the checkpoint and all its conditioning components, resolution, batch size, numerical precision, guidance scale, and any negative prompt. Use the same scheduler family, which chooses the noise levels and the rule for updating the sample. Use a deterministic mode that both settings support and fix the rule for spacing the steps. Record the actual schedule for each run, since the 10-step and 30-step runs visit different levels. With deterministic updates, each result is determined by its initial tensor and conditioning. Record both the step count and the number of denoiser evaluations. Some sampling or guidance implementations evaluate the network more than once per step or combine multiple conditions into one batch.

Measure **denoising time** from the start of the first sampler step, with the initial tensor and conditioning ready on device, to the end of the last update, with the device synchronized. Also measure **end-to-end generation time** from an input prompt ready for encoding to a decoded image ready to display. Include text encoding and image decoding in that measure, and state whether transfer to the host is included. Keep model loading and file saving outside both measurements unless measuring them explicitly. Warm up each setting on the same hardware, alternate run order, synchronize accelerator work at the timing boundaries, and report median and tail latency over repeated runs.

Score quality on blinded pairs in a randomized order. Check **prompt adherence** against a checklist, such as the two shapes, both colors, and the above/below relation. Score **visual integrity** separately, for example whether edges are coherent and whether the image contains artifacts or malformed shapes. A sharp image with the colors swapped can do well on integrity and fail adherence. If the task values varied outputs, measure diversity across different starting tensors for each prompt under each setting, and report it separately. Keep the paired scores and raw counts. The comparison describes the effect of the two schedules for this checkpoint and setup.

10. Check motion as well as appearance

Build three groups of prompts with similar clip lengths and fixed frame sampling. Include details that make the intended motion observable. A perfectly uniform sphere, for example, would make a rotation hard to verify.

Prompt	Check that uses more than one frame
A white cube with a blue dot on its front face and a red stripe on its right face rotates on a table. The camera stays still.	Compare early, middle, and late frames. The visible faces and the positions of the dot and stripe should change consistently with one rotating cube, while fixed background landmarks stay in place.
A camera moves sideways without turning as it films a stationary statue, a near post, and a distant wall.	Track the post and wall across frames. A sideways camera move can produce parallax, a larger image displacement for a nearer landmark than for a farther one. Check that the view changes without the statue walking or changing shape.
A stationary camera films a still life with a green mug and an apple on a table. Nothing moves.	Compare the mug, apple, and table corners across sampled frames. Positions and silhouettes should stay stable. Also check for visible color or texture flicker across adjacent frames.

Freeze the prompt text, expected motion, clip length, frame sampling, and scoring rules before looking at any outputs. Use the same temporal sampling for systems under comparison. For a qualitative check, give reviewers the same short clip and a contact sheet with the same labeled frame times. Label a result as unclear when the relevant landmark is occluded throughout or the frames cannot establish the intended motion. Record such cases instead of treating them as successes.

For every clip, record two fields. The **appearance** field checks whether the requested objects and attributes are present. The **motion** field checks the temporal requirement for that prompt. For example, if a cube has the right markers but they slide across an unchanged face, record appearance correct and motion wrong. If a camera is meant to move but the output repeats the same frame, the appearance can still be correct. Report the counts for both checks, including appearance-correct/motion-wrong cases, and break them down by motion group. This makes it possible to see whether a generator draws the scene well while missing its dynamics.

11. Separate variety from prompt following

Choose prompts that specify some content and leave room for different compositions. For example, *two yellow cups and a blue plate on a table* fixes object counts and colors while allowing different viewpoints, crops, and arrangements. For each prompt, save several starting noise tensors and generate at $g = 1$ and $g = 2$ in the chapter's guidance convention with every tensor. Use the same deterministic sampler at both scales. Keep the checkpoint, schedule, step count, resolution, and remaining sampling settings fixed. This pairs the two results for each prompt and draw without requiring different draws to look alike.

Define prompt adherence before seeing the images. In the example, check whether there are two cups, whether they are yellow, and whether there is a blue plate on the table. Record the fraction of outputs satisfying all requirements as well as the individual checks. Compare guidance settings on the same prompt–draw pairs. This shows whether higher guidance helps particular requirements, such as the object count, without folding that effect into a diversity score.

Within each prompt and scale, inspect how different draws arrange the scene. Define composition categories such as overhead or side view, tight or wide crop, and object placement relative to the center. Use blinded reviewers and a fixed rule. Count a pair as repeated when the viewpoint, crop, and placement labels all agree. Record pairs that cannot be scored separately and state how many pairs are scorable. With eight scorable draws there are $8 \cdot 7/2 = 28$ unordered pairs. If six pairs repeat the same composition, the repeat fraction is $6/28$. Compute that fraction for each prompt before averaging so a prompt with more draws does not dominate. If you also compute the fraction only among outputs that follow the prompt, report the number of qualifying outputs and pairs. A setting with very few qualifying draws provides little evidence about diversity among successful ones.

Saving only the strongest sample per prompt changes the experiment to **best of several draws**. An adherence-based selector can hide the rate of weak or incorrect generations. One saved image also leaves no within-prompt pairs from which to measure repeated composition. If best-of-several performance is part of the study, fix the selection rule and give both scales the same number of attempts. Keep all generations so you can report the single-draw adherence and variety as well as the selected-sample results.